

Evaluating Haptic and Audio Cues in Complex Pathfinding Tasks

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Maze navigation serves as a simplified model of a pathfinding task that captures spatial planning, coordination, and error correction while allowing precise measurement of speed and accuracy. In engineering and user interface design, subjects often rely on real-time feedback in guidance of their action, the specific modality of which can significantly impact their performance. This project investigates the effect of visual and haptic feedback, in the forms of flashing light and tactile vibration respectively, on the speed and accuracy of a maze-solving task.

1 Hypothesis

Haptic feedback, in the form of a phone vibration, will allow subjects to complete the maze in the least amount of time and with the least number of times hitting the wall, achieving the greatest speed with the highest accuracy.

There will be a negative correlation between speed and accuracy, meaning that as the speed of the subject to complete the maze increases, the accuracy decreases.

2 Experimental Design

The experiment begins by presenting a pre-generated Python 2D maze. Subjects will be tasked to maneuver a marker through the maze using the “WASD” keys on a keyboard.

Each subject will be assigned one of two random feedback types, which trigger when the marker contacts the wall: 1) a phone vibration (haptic); and 2) a red screen flash (audio). Both of these are classified as immediate and simple.

The accuracy will be measured as the total time, in seconds the player is in contact with

the maze walls. This timer will be implemented through the maze program.

The speed will be measured as the total time, in seconds, it takes for the subject to traverse from the start to the end of the maze. The timer will automatically begin as the marker crosses the entrance, and stop when it reaches the exit.

3 Data Collection

24 subjects will be evaluated for each of the 2 feedback types, for a total of 48 subjects. These students will be selected from Year 2 Engineering Science students, due to their similar ages, experiences, and computer competency. They will be selected using simple random sampling, by drawing names from a list of students from the “People” tab on Quercus. This aims to eliminate convenience sampling.

Identical instructions will be given to each subject before the start of each trial with a scripted protocol to ensure consistency across all participants. Furthermore, each participant will be given the same laptop, as well as vibration and audio intensity. Tests will be conducted in the same setting with limited distractions.

4 Data Analysis

Using R/RStudio, the mean, median, standard deviation, and interquartile range will be computed. In addition, a box plot and a scatter plot will be generated, which will identify outliers, allow inspection of distributions, and analyze the spread of data collected.

Then, the data will be tested for normality using Q-Q plots. To compare the task completion speed and accuracy between the two distinct feedback types, independent samples t-tests will be conducted

Furthermore, the Pearson correlation coefficient will be computed to quantify and verify the relation between speed and accuracy that was hypothesized. All statistical tests will be evaluated using a significance level of $\alpha = 0.05$.